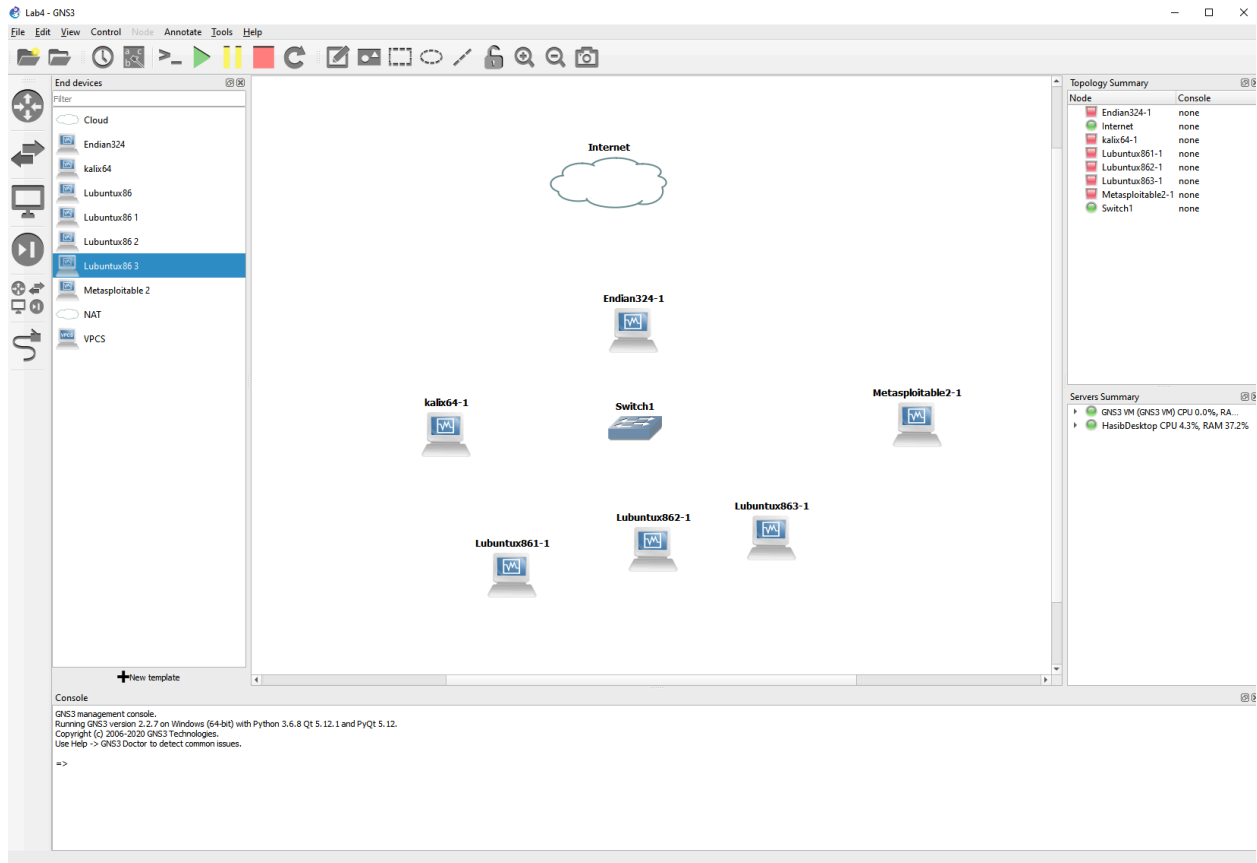


Lab 4

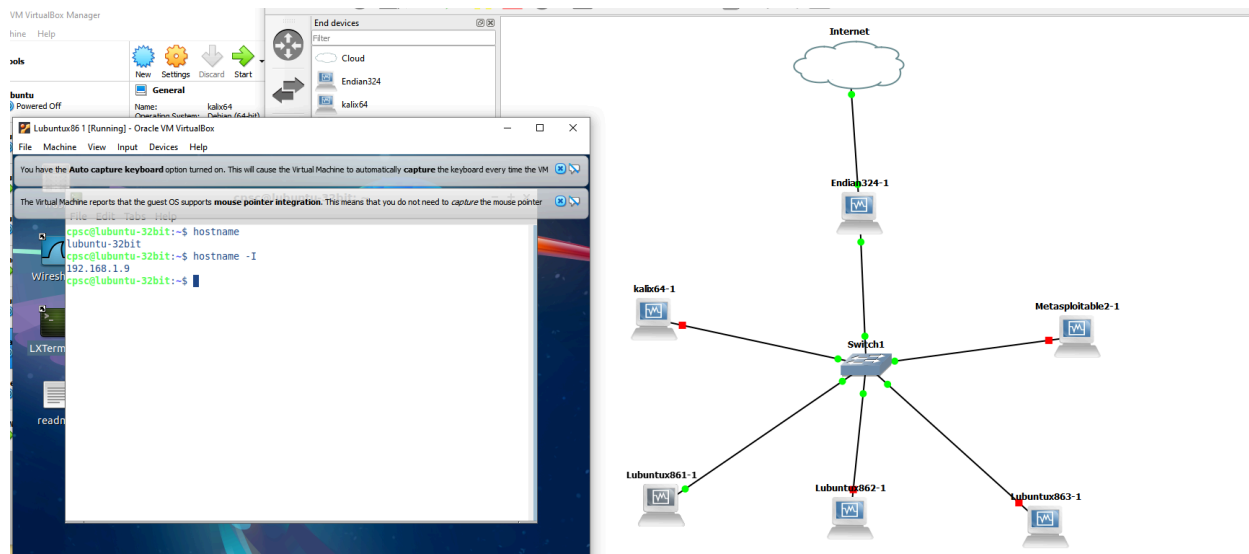


Adding in the different VM's to create the Topology
(Steps 1 – 4)

5. Attach an Ethernet link to connect Endian's eth0 interface to an available switch port. Create a table to match VM interface to the used switch port.

adapter:	Internet	Endian	Metaspl.	lubuntu1	lubuntu2	lubuntu3	Kali	Switch
Eth0	Endian	Switch1	Endian	Endian	Endian	Endian	Endian	Endian
Eth1		Internet						Meta
Eth2								lubuntu1
Eth3								lubuntu2
Eth4								lubuntu3
Eth5								Kali

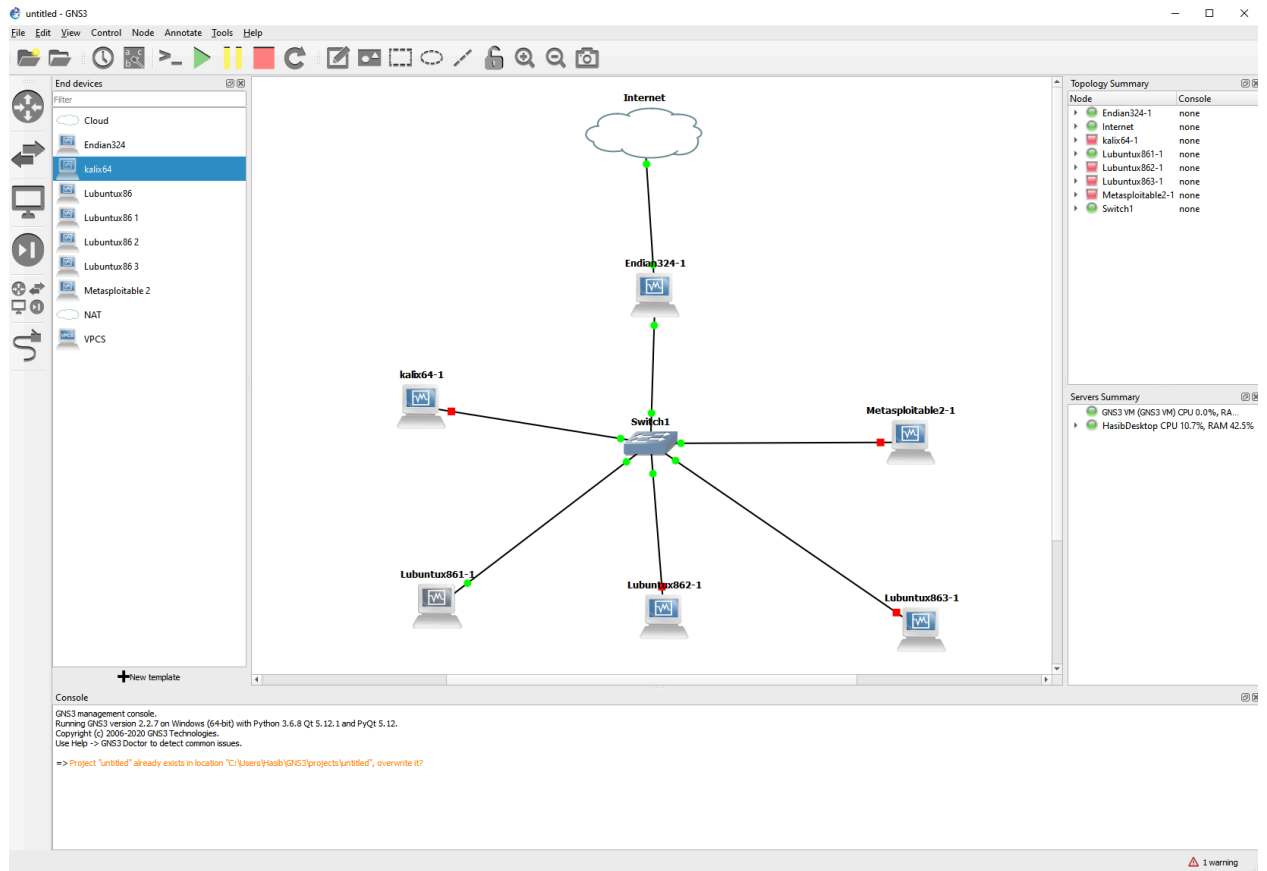
Steps 6-10:



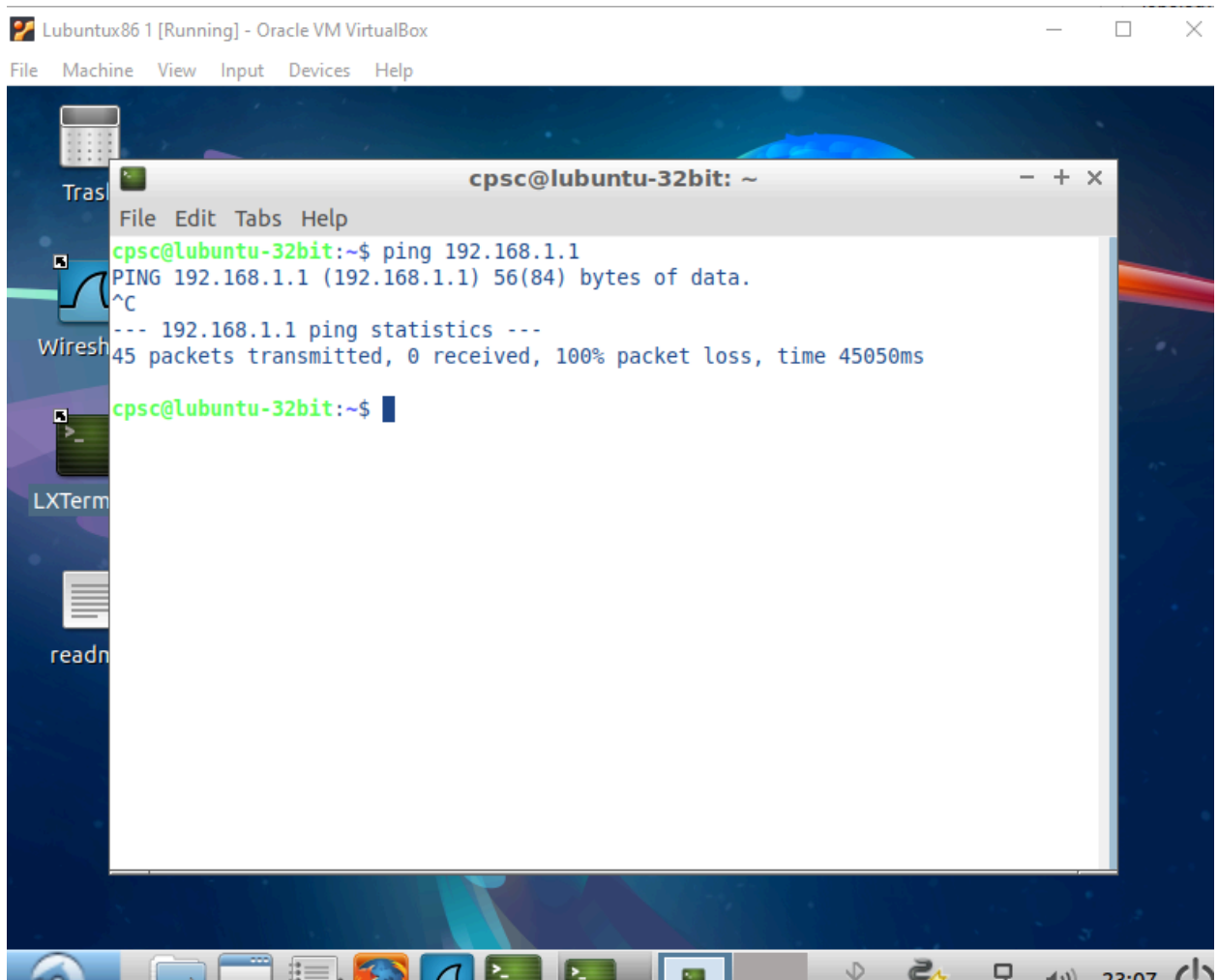
The given IP address is 192.168.1.9

At this point, I was not getting any connection on my lUbuntu VM and was not able to ping the Endian router, nor google.

I restarted the lab a third time here to try to figure out the configurations issues I was running into.



A second attempt is made to ping the Endian router



```
Lubuntu86 1 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

cpsec@lubuntu-32bit: ~
File Edit Tabs Help
cpsec@lubuntu-32bit:~$ ping 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
^C
--- 192.168.1.1 ping statistics ---
45 packets transmitted, 0 received, 100% packet loss, time 45050ms

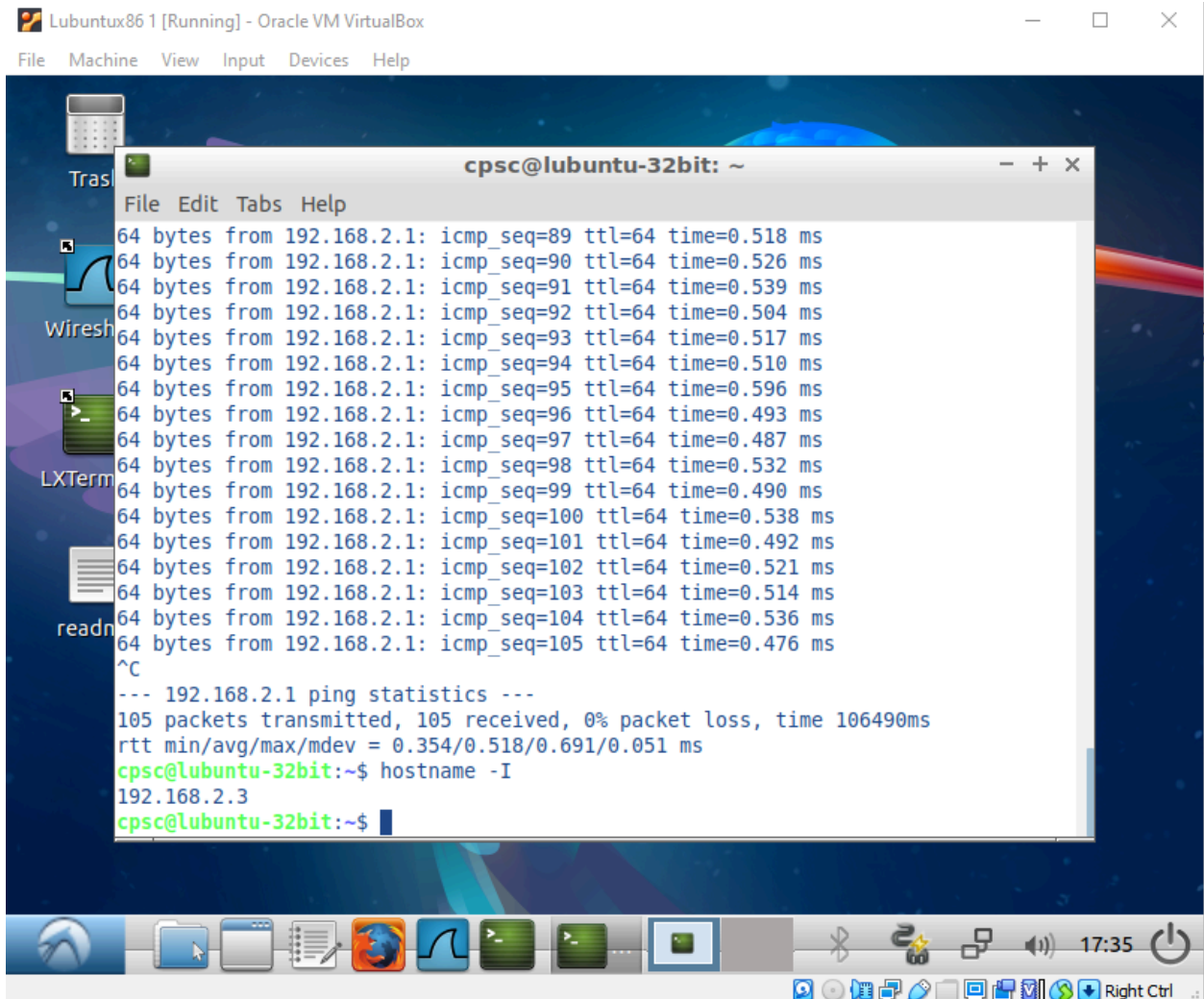
cpsec@lubuntu-32bit:~$
```

This was the same issue I ran into last time. At this point, I believe that I have set up everything incorrectly.

This problem was solved, and we are now, finally after a few days, able to continue the Lab

After the Zoom Meeting, we were able to determine that the DHCP server on Endian was not enabled! Whoops.

#10)



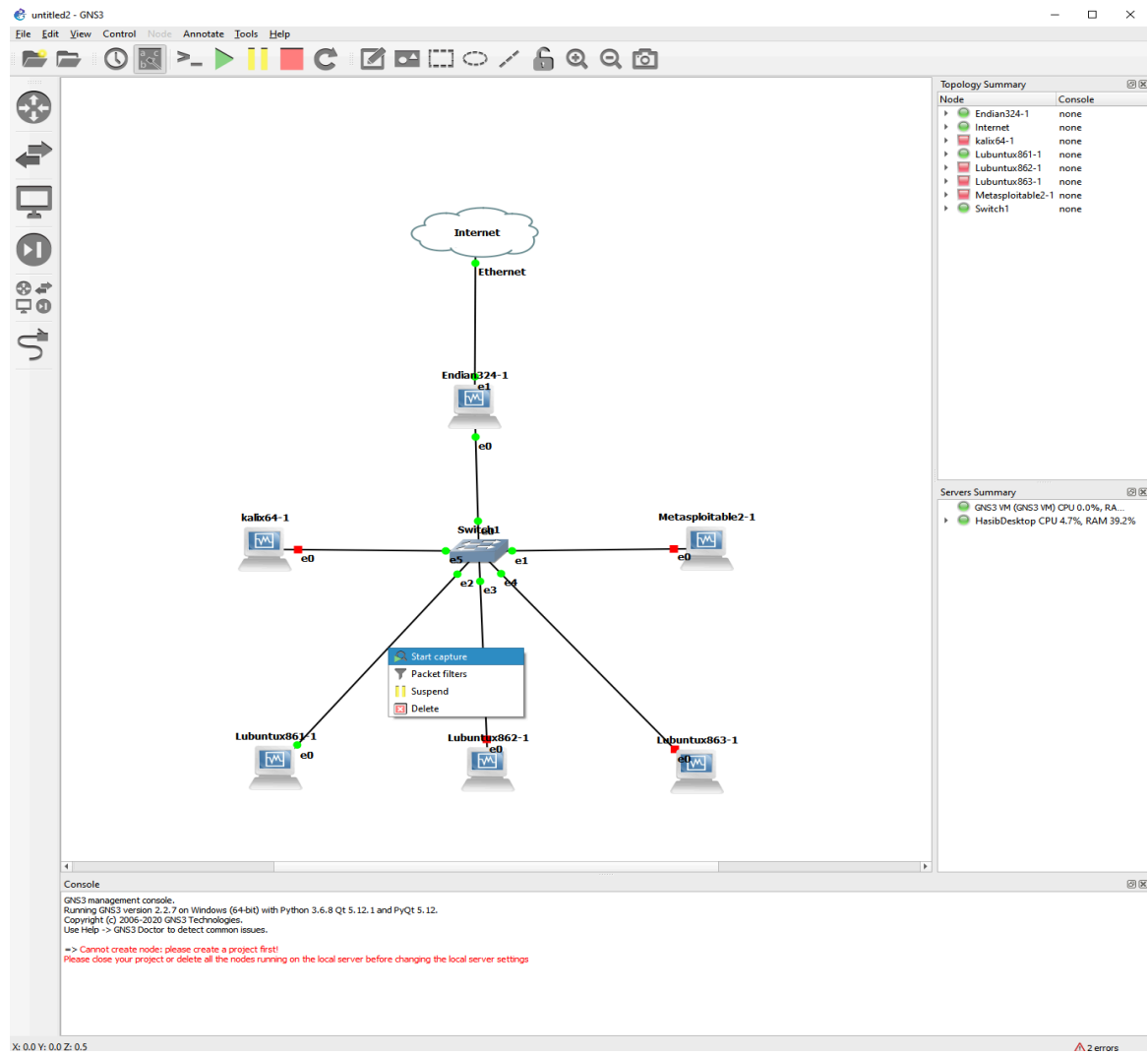
```
Lubuntu 18.04 LTS [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

cpssc@lubuntu-32bit: ~
File Edit Tabs Help
64 bytes from 192.168.2.1: icmp_seq=89 ttl=64 time=0.518 ms
64 bytes from 192.168.2.1: icmp_seq=90 ttl=64 time=0.526 ms
64 bytes from 192.168.2.1: icmp_seq=91 ttl=64 time=0.539 ms
64 bytes from 192.168.2.1: icmp_seq=92 ttl=64 time=0.504 ms
64 bytes from 192.168.2.1: icmp_seq=93 ttl=64 time=0.517 ms
64 bytes from 192.168.2.1: icmp_seq=94 ttl=64 time=0.510 ms
64 bytes from 192.168.2.1: icmp_seq=95 ttl=64 time=0.596 ms
64 bytes from 192.168.2.1: icmp_seq=96 ttl=64 time=0.493 ms
64 bytes from 192.168.2.1: icmp_seq=97 ttl=64 time=0.487 ms
64 bytes from 192.168.2.1: icmp_seq=98 ttl=64 time=0.532 ms
64 bytes from 192.168.2.1: icmp_seq=99 ttl=64 time=0.490 ms
64 bytes from 192.168.2.1: icmp_seq=100 ttl=64 time=0.538 ms
64 bytes from 192.168.2.1: icmp_seq=101 ttl=64 time=0.492 ms
64 bytes from 192.168.2.1: icmp_seq=102 ttl=64 time=0.521 ms
64 bytes from 192.168.2.1: icmp_seq=103 ttl=64 time=0.514 ms
64 bytes from 192.168.2.1: icmp_seq=104 ttl=64 time=0.536 ms
64 bytes from 192.168.2.1: icmp_seq=105 ttl=64 time=0.476 ms
^C
--- 192.168.2.1 ping statistics ---
105 packets transmitted, 105 received, 0% packet loss, time 106490ms
rtt min/avg/max/mdev = 0.354/0.518/0.691/0.051 ms
cpssc@lubuntu-32bit:~$ hostname -I
192.168.2.3
cpssc@lubuntu-32bit:~$
```

The given IPv4 address for this machine is 192.168.2.3

As can be seen in the screenshot, we were able to ping Endian (192.168.2.1) and google.com

#11)



You can also use this command:

```
dhclient -r interface
```

Where `interface` is the device you want to get a new address for.

```
dhclient -r eth0
```

The `-r` flag forces `dhclient` to first release any leases you have, you can then use this command to request a new lease:

```
dhclient eth0
```

From `man dhclient`:

```
-r      Tell dhclient to release the current lease it has from the
        server.  This is not required by the DHCP protocol, but some
        ISPs require their clients to notify the server if they wish
        to release an assigned IP address.
```

----- This section follows -----

----- This section follows -----

----- This section follows -----

Capturing from - [Lubuntu861-1 Ethernet0 to Switch1 Ethernet2]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
19	118.729397	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8656f60c
20	118.729397	192.168.2.1	192.168.2.3	ICMP	62	Echo (ping) request id=0xc072, seq=0/0, ttl=
21	119.729796	192.168.2.1	192.168.2.3	DHCP	342	DHCP Offer - Transaction ID 0x8656f60c
22	119.729796	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Transaction ID 0x8656f60c
23	119.745412	192.168.2.1	192.168.2.3	DHCP	342	DHCP ACK - Transaction ID 0x8656f60c
24	119.763957	192.168.2.3	224.0.0.22	IGMPv3	60	Membership Report / Join group 224.0.0.251 fo
25	119.854724	192.168.2.3	224.0.0.251	MDNS	87	Standard query 0x0000 PTR _ipps._tcp.local, "
26	119.953300	192.168.2.3	224.0.0.251	MDNS	255	Standard query 0x0000 ANY 5.9.3.9.f.2.e.1.2.2
27	120.203155	192.168.2.3	224.0.0.251	MDNS	255	Standard query 0x0000 ANY 5.9.3.9.f.2.e.1.2.2
28	120.216820	192.168.2.3	224.0.0.22	IGMPv3	60	Membership Report / Join group 224.0.0.251 fo
29	120.454963	192.168.2.3	224.0.0.251	MDNS	255	Standard query 0x0000 ANY 5.9.3.9.f.2.e.1.2.2
30	120.655043	192.168.2.3	224.0.0.251	MDNS	237	Standard query response 0x0000 PTR, cache flu
31	120.854147	192.168.2.3	224.0.0.251	MDNS	87	Standard query 0x0000 PTR _ipps._tcp.local, "
32	121.804771	192.168.2.3	224.0.0.251	MDNS	237	Standard query response 0x0000 PTR, cache flu
33	122.855923	192.168.2.3	224.0.0.251	MDNS	87	Standard query 0x0000 PTR _ipps._tcp.local, "
34	123.675763	PcsCompu_d0:88:66	Broadcast	ARP	60	Who has 192.168.2.1? Tell 192.168.2.3
35	123.675763	PcsCompu_e9:4b:51	PcsCompu_d0:88:66	ARP	60	192.168.2.1 is at 08:00:27:e9:4b:51
36	123.675763	192.168.2.3	192.168.2.1	DNS	87	Standard query 0xba49 AAAA daisy.ubuntu.com O
37	123.676739	192.168.2.1	192.168.2.3	DNS	87	Standard query response 0xba49 AAAA daisy.ubu
38	123.680644	192.168.2.3	192.168.2.1	DNS	87	Standard query 0x1d0e A daisy.ubuntu.com OPT
39	123.680644	192.168.2.3	192.168.2.1	DNS	87	Standard query 0xd6f1 AAAA daisy.ubuntu.com O
40	123.681619	192.168.2.1	192.168.2.3	DNS	87	Standard query response 0xd6f1 AAAA daisy.ubu
41	123.691379	192.168.2.1	192.168.2.3	DNS	135	Standard query response 0x1d0e A daisy.ubuntu
42	123.954899	192.168.2.3	224.0.0.251	MDNS	237	Standard query response 0x0000 PTR, cache flu
43	125.386690	192.168.2.3	192.168.2.1	DNS	87	Standard query 0x2a79 AAAA daisy.ubuntu.com O
44	125.387666	192.168.2.1	192.168.2.3	DNS	87	Standard query response 0x2a79 AAAA daisy.ubu
45	126.858498	192.168.2.3	224.0.0.251	MDNS	87	Standard query 0x0000 PTR _ipps._tcp.local, "
46	128.678738	PcsCompu_e9:4b:51	PcsCompu_d0:88:66	ARP	60	Who has 192.168.2.3? Tell 192.168.2.1
47	128.678738	PcsCompu_d0:88:66	PcsCompu_e9:4b:51	ARP	60	192.168.2.3 is at 08:00:27:d0:88:66
48	134.861696	192.168.2.3	224.0.0.251	MDNS	87	Standard query 0x0000 PTR _ipps._tcp.local, "

< Frame 1: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface 0
 > Ethernet II, Src: PcsCompu_d0:88:66 (08:00:27:d0:88:66), Dst: 255.255.255.255
 > Internet Protocol Version 4, Src: 192.168.1.9, Dst: 255.255.255.255
 > User Datagram Protocol, Src Port: 68, Dst Port: 67
 > Dynamic Host Configuration Protocol (Request)

0000 ff ff ff ff ff 08 00 27 d0 88 66 08 00 45 11
 0010 01 48 00 00 00 00 80 11 77 e4 c0 a8 01 09 ff
 0020 ff ff 00 44 00 43 01 34 06 a0 01 01 06 00 aa
 0030 ab 73 06 d9 00 00 c0 a8 01 09 00 00 00 00 00

Ready to load or capture

Lubuntu86 1 [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

cpssc@lubuntu-32bit: ~

File Edit Tabs Help

ether 08:00:27:d0:88:66 txqueuelen 1000 (Ethernet)
 RX packets 1019 bytes 90459 (90.4 KB)
 RX errors 0 dropped 0 overruns 0 frame 0
 TX packets 2721 bytes 251695 (251.6 KB)
 TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

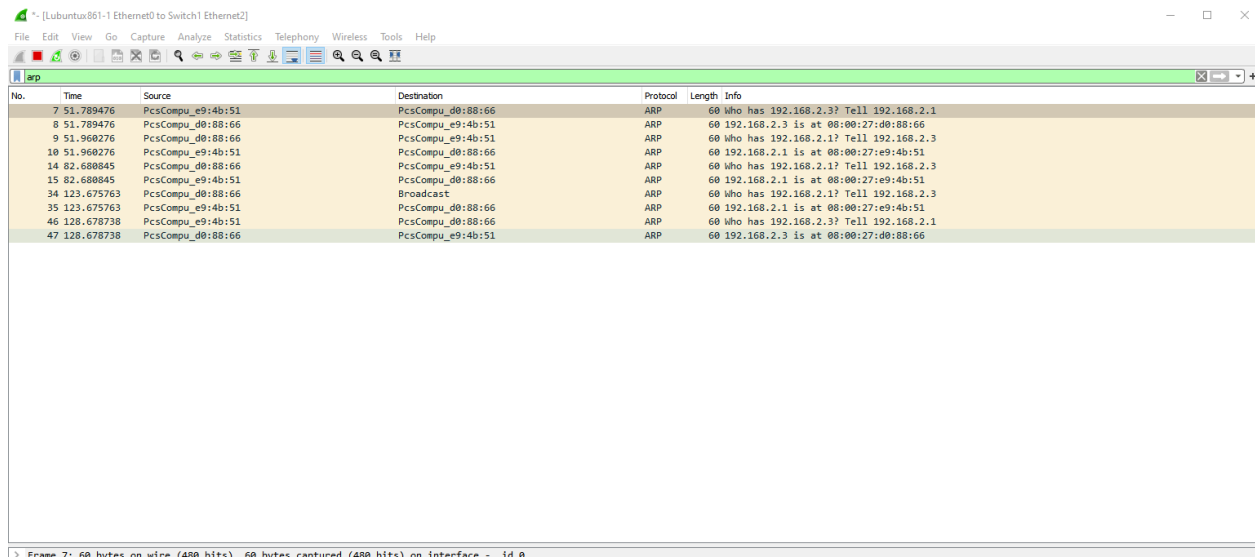
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
 inet 127.0.0.1 netmask 255.0.0.0
 inet6 ::1 prefixlen 128 scopeid 0x10<host>
 loop txqueuelen 1000 (Local Loopback)
 RX packets 3193 bytes 249196 (249.1 KB)
 RX errors 0 dropped 0 overruns 0 frame 0
 TX packets 3193 bytes 249196 (249.1 KB)
 TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

cpssc@lubuntu-32bit:~\$ dhclient -r enp0s3
 cpssc@lubuntu-32bit:~\$ dhclient enp0s3
 RTNETLINK answers: Operation not permitted
 cpssc@lubuntu-32bit:~\$ sudo dhclient -r enp0s3
 cpssc@lubuntu-32bit:~\$ dhclient enp0s3
 RTNETLINK answers: Operation not permitted
 cpssc@lubuntu-32bit:~\$ sudo dhclient enp0s3
 cmp: EOF on /tmp/fusneYCXK which is empty
 cpssc@lubuntu-32bit:~\$

17:46

#12-14)

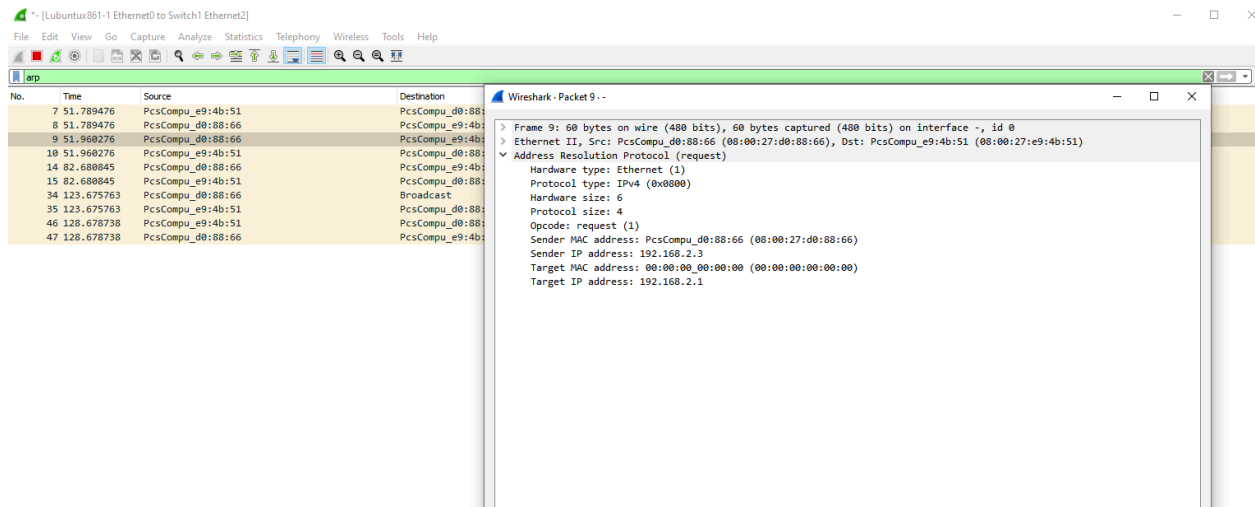
After running the command, we were able to filter out the ARP packets like so:



The screenshot shows the Wireshark interface with the filter 'arp' applied. The packet list table contains 13 entries, all of which are ARP packets. The status bar at the bottom indicates that 60 bytes were captured (480 bits) on interface 'id 0'.

No.	Time	Source	Destination	Protocol	Length	Info
7	51.789476	PcsCompu_e9:4b:51	PcsCompu_d0:88:66	ARP	60	Who has 192.168.2.3? Tell 192.168.2.1
8	51.789476	PcsCompu_d0:88:66	PcsCompu_e9:4b:51	ARP	60	192.168.2.3 is at 08:00:27:d0:88:66
9	51.960276	PcsCompu_d0:88:66	PcsCompu_e9:4b:51	ARP	60	Who has 192.168.2.1? Tell 192.168.2.3
10	51.960276	PcsCompu_e9:4b:51	PcsCompu_d0:88:66	ARP	60	192.168.2.1 is at 08:00:27:e9:4b:51
14	82.680845	PcsCompu_d0:88:66	PcsCompu_e9:4b:51	ARP	60	Who has 192.168.2.1? Tell 192.168.2.3
15	82.680845	PcsCompu_e9:4b:51	PcsCompu_d0:88:66	ARP	60	192.168.2.1 is at 08:00:27:e9:4b:51
34	123.675763	PcsCompu_d0:88:66	Broadcast	ARP	60	Who has 192.168.2.1? Tell 192.168.2.3
35	123.675763	PcsCompu_e9:4b:51	PcsCompu_d0:88:66	ARP	60	192.168.2.1 is at 08:00:27:e9:4b:51
46	128.678738	PcsCompu_e9:4b:51	PcsCompu_d0:88:66	ARP	60	Who has 192.168.2.3? Tell 192.168.2.1
47	128.678738	PcsCompu_d0:88:66	PcsCompu_e9:4b:51	ARP	60	192.168.2.3 is at 08:00:27:d0:88:66

This screenshot shows an ARP request message.



The screenshot shows the Wireshark interface with the filter 'arp' applied. The packet list table contains 13 entries, all of which are ARP packets. The packet details pane on the right shows the structure of the selected packet (Frame 9).

No.	Time	Source	Destination
7	51.789476	PcsCompu_e9:4b:51	PcsCompu_d0:88:66
8	51.789476	PcsCompu_d0:88:66	PcsCompu_e9:4b:51
9	51.960276	PcsCompu_d0:88:66	PcsCompu_e9:4b:51
10	51.960276	PcsCompu_e9:4b:51	PcsCompu_d0:88:66
14	82.680845	PcsCompu_d0:88:66	PcsCompu_e9:4b:51
15	82.680845	PcsCompu_e9:4b:51	PcsCompu_d0:88:66
34	123.675763	PcsCompu_d0:88:66	Broadcast
35	123.675763	PcsCompu_e9:4b:51	PcsCompu_d0:88:66
46	128.678738	PcsCompu_e9:4b:51	PcsCompu_d0:88:66
47	128.678738	PcsCompu_d0:88:66	PcsCompu_e9:4b:51

Wireshark - Packet 9 --

- > Frame 9: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
- > Ethernet II, Src: PcsCompu_d0:88:66 (08:00:27:d0:88:66), Dst: PcsCompu_e9:4b:51 (08:00:27:e9:4b:51)
- > Address Resolution Protocol (request)
 - Hardware type: Ethernet (1)
 - Protocol type: IPv4 (0x0800)
 - Hardware size: 6
 - Protocol size: 4
 - Opcode: request (1)
 - Sender MAC address: PcsCompu_d0:88:66 (08:00:27:d0:88:66)
 - Sender IP address: 192.168.2.3
 - Target MAC address: 00:00:00:00:00:00 (00:00:00:00:00:00)
 - Target IP address: 192.168.2.1

As can be seen, the source address is 08:00:27:d0:88:66 , I believe this is the MAC address for the 'NIC' of the lUbuntu1 VM. The destination for this message is 00:00:00:00:00:00, which signifies that it is a broadcast.

#15)

The ARP reply message can be seen below:

The screenshot shows a Wireshark packet capture of ARP traffic. The main window displays a list of packets, with packet 10 selected. The packet list shows the following details:

No.	Time	Source	Destination	Protocol	Length	Info
7	51.789476	PcsCompu_e9:4b:51	PcsCompu_d0:88:66	ARP	60	Who has 192.168.2.3? Tell 192.168.2.1
8	51.789476	PcsCompu_d0:88:66	PcsCompu_e9:4b:51	ARP	60	192.168.2.3 is at 08:00:27:d0:88:66
9	51.968276	PcsCompu_d0:88:66	PcsCompu_e9:4b:51	ARP	60	Who has 192.168.2.1? Tell 192.168.2.3
10	51.968276	PcsCompu_e9:4b:51	PcsCompu_d0:88:66	ARP	60	192.168.2.1 is at 08:00:27:e9:4b:51
14	82.680845	PcsCompu_d0:88:66	PcsCompu_e9:4b:51	ARP	60	Who has 192.168.2.1? Tell 192.168.2.3
15	82.680845	PcsCompu_e9:4b:51	PcsCompu_d0:88:66	ARP	60	192.168.2.1 is at 08:00:27:e9:4b:51
34	123.675763	PcsCompu_d0:88:66	Broadcast	ARP	60	Who has 192.168.2.1? Tell 192.168.2.3
35	123.675763	PcsCompu_e9:4b:51	PcsCompu_d0:88:66	ARP	60	192.168.2.1 is at 08:00:27:e9:4b:51
46	128.678738	PcsCompu_e9:4b:51	PcsCompu_d0:88:66	ARP	60	Who has 192.168.2.3? Tell 192.168.2.1
47	128.678738	PcsCompu_d0:88:66	PcsCompu_e9:4b:51	ARP	60	192.168.2.3 is at 08:00:27:d0:88:66

The detailed view of packet 10 shows the following structure:

```
> Frame 10: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
> Ethernet II, Src: PcsCompu_e9:4b:51 (08:00:27:e9:4b:51), Dst: PcsCompu_d0:88:66 (08:00:27:d0:88:66)
  Address Resolution Protocol (reply)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0000)
    Hardware size: 6
    Protocol size: 4
    Opcode: reply (2)
    Sender MAC address: PcsCompu_e9:4b:51 (08:00:27:e9:4b:51)
    Sender IP address: 192.168.2.1
    Target MAC address: PcsCompu_d0:88:66 (08:00:27:d0:88:66)
    Target IP address: 192.168.2.3
```

The source address shown here is 08:00:27:e9:4b:51, which I believe would be the MAC address for the 'NIC' of the Endian VM. The destination is the same address in the previous screenshot, 08:00:27:d0:88:66, and we can see from this exchange that it is the Endian router responding to the ARP request from the IUbuntu VM.